Group Topic: PH Staple Food Price Analytics — SDG 2 Lab No. Chapter 8

Git Repo/ Colab Link: https://github.com/jaysonbuenosaires/Buenosaires_Elec2 Date:May 11, 2026

Objectives

- To process a large-scale dataset of Philippine food prices (2.5 million rows) and environmental data to identify price drivers.
- To implement advanced feature engineering including rolling weather averages and price momentum lags.
- To develop a Random Forest model for price prediction and categorize regions based on food security risk levels.

Tasks Completed

Lead ML Engineer

- **Tasks:** Model development and training.
- **Description:**
  - Implemented a Time-Series Split to prevent data leakage, using the final 20% of sorted rows (approximately late 2020 to mid-2021) as the test set, with a guaranteed minimum of 50 rows.
  - Developed and trained a baseline Random Forest Regressor on a 700,000-row sample using n_jobs=-1 for parallel processing and computational efficiency.
  - Configured baseline hyper-parameters: 20 estimators, max_depth=8, and min_samples_leaf=20, achieving an initial MAE and RMSE on the held-out test set.
- **Screenshot/Code Snippet:**

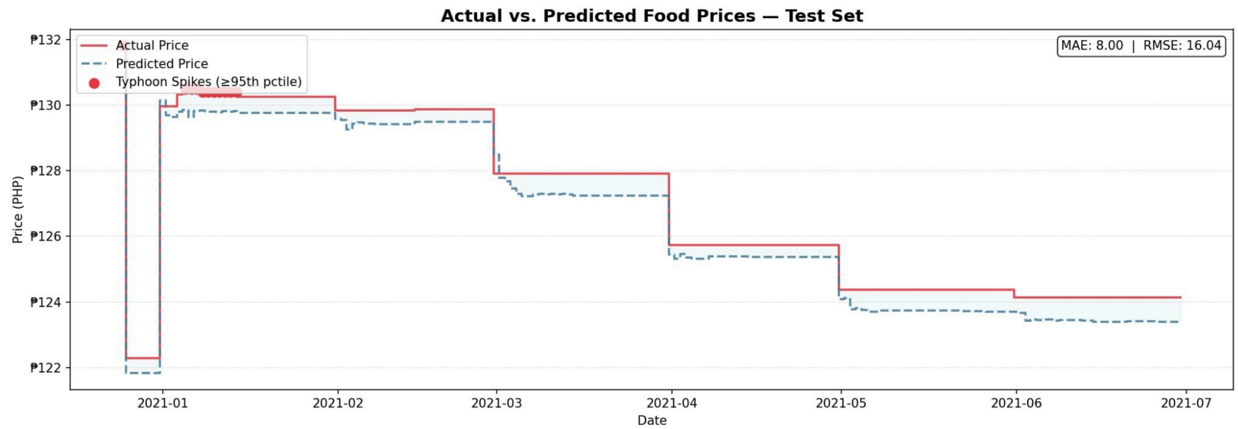
```python
# ── 2-B Train baseline Random Forest (FAST VERSION) ────────

sample_size = 700_000   # start smaller for testing

train_sample = train_df.sample(
    n=min(sample_size, len(train_df)),
    random_state=42
)

X_train_small = train_sample[FEATURE_COLS]
y_train_small = train_sample[TARGET_COL]

print("Training rows:", len(X_train_small))

rf_baseline = RandomForestRegressor(
    n_estimators=20,
    max_depth=8,
    min_samples_leaf=20,
    random_state=42,
    n_jobs=-1
)

rf_baseline.fit(X_train_small, y_train_small)

y_pred_baseline = rf_baseline.predict(X_test)

mae_b = mean_absolute_error(y_test, y_pred_baseline)
rmse_b = np.sqrt(mean_squared_error(y_test, y_pred_baseline))

print(f'Baseline RF → MAE: {mae_b:.4f} | RMSE: {rmse_b:.4f}')
```

Tasks to be Completed

- Conduct a Grid Search Cross-Validation to optimize the max_depth and n_estimators of the Random Forest model.

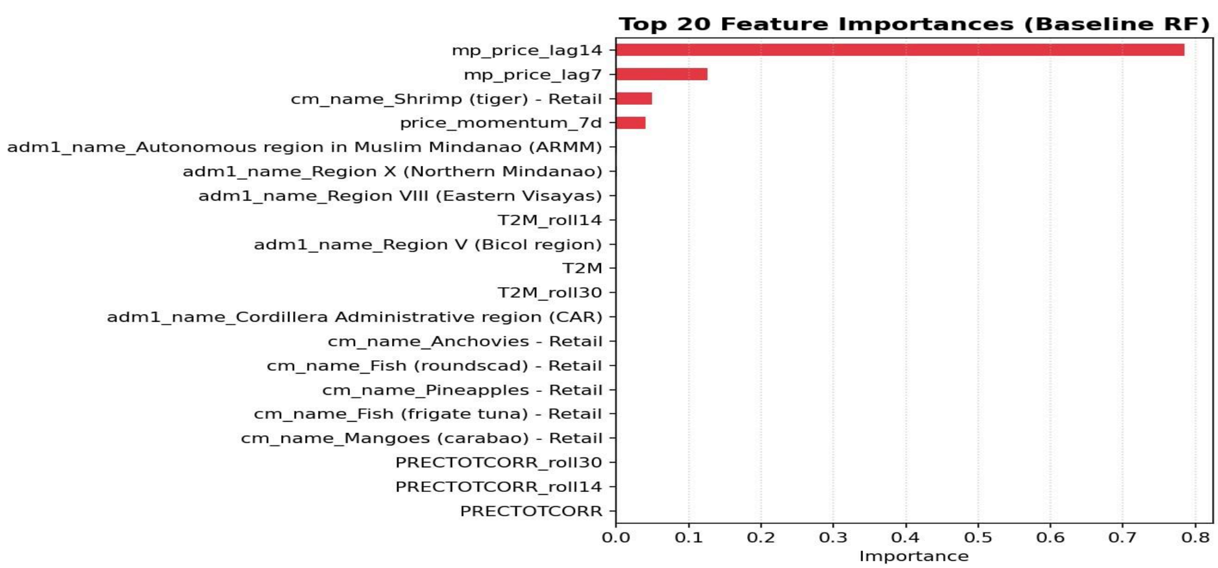
Results and Analysis

1. Price Prediction — Baseline Random Forest



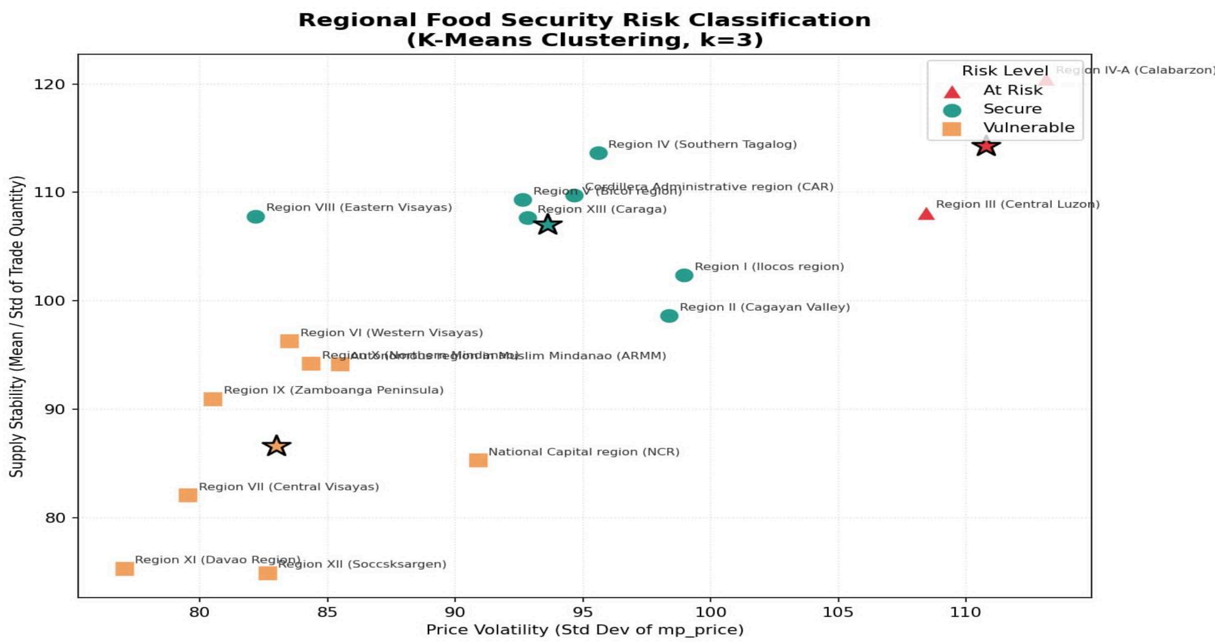
The Baseline Random Forest model was tested on data covering January to July 2021, achieving a MAE of ₱8.00 and RMSE of ₱16.04. As shown in the figure, the predicted price (dashed blue) generally follows the trend of the actual price (solid red) but consistently lags behind during sharp price transitions — particularly during the early January 2021 spike and the step-down drops in March and April 2021. This lag indicates that the baseline model underreacts to sudden market changes driven by typhoon events or supply disruptions.

2. Feature Importance Analysis



The feature importance chart reveals that mp_price_lag14 is by far the most dominant predictor (~0.78 importance score), followed by mp_price_lag7 (~0.12) and price_momentum_7d. This confirms that recent price history is the strongest signal for forecasting future prices. Environmental variables such as T2M (2-meter temperature) and its rolling averages (T2M_roll14, T2M_roll30), along with precipitation features (PRECTOTCORR, PRECTOTCORR_roll14, PRECTOTCORR_roll30), also appear in the top 20, validating that climate data meaningfully contributes to price prediction. Regional indicators (e.g., ARMM, Region X, Region VIII) and specific commodities (e.g., Shrimp, Anchovies, Mangoes) further contribute, reflecting localized price behavior across the Philippines.

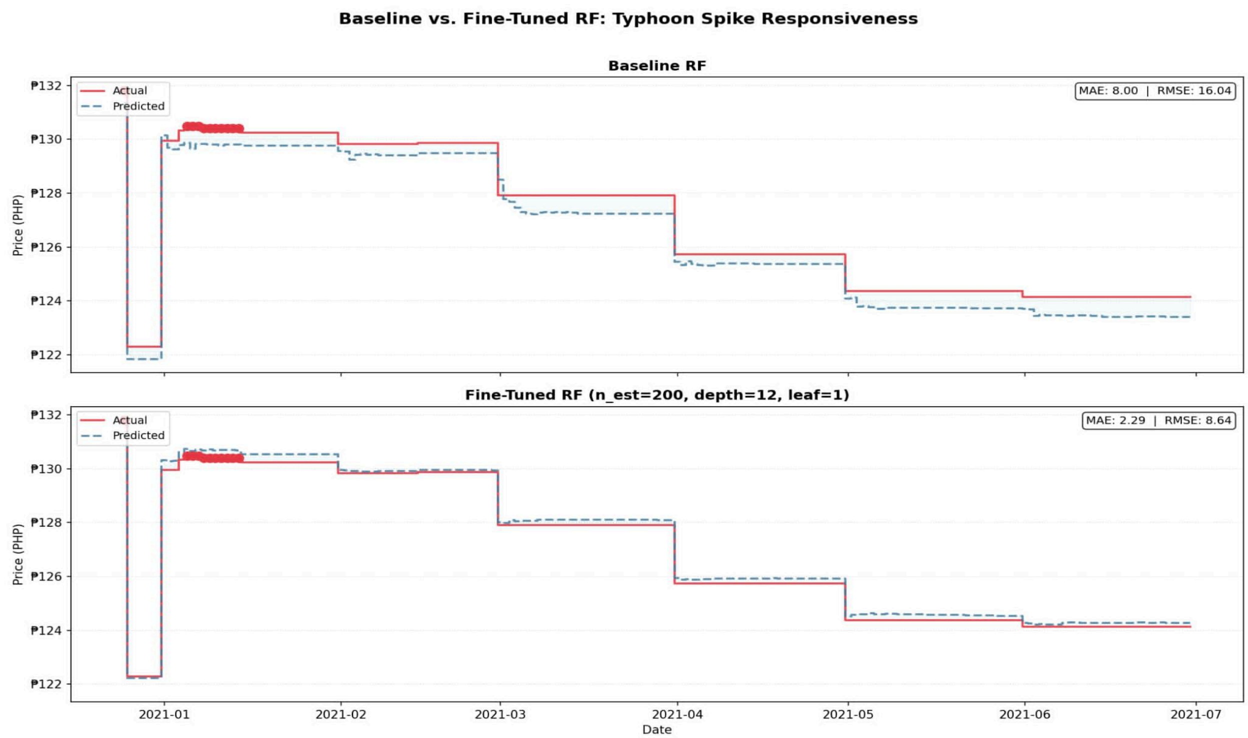
3. Regional Food Security Risk Classification — K-Means Clustering (k=3)



Using K-Means clustering on price volatility and supply stability, all 17 Philippine regions were categorized into three risk levels:

- At Risk — Region III (Central Luzon) and Region IV-A (Calabarzon) recorded the highest price volatility scores, driven by heavy urban demand and their role as major food distribution hubs.
- Vulnerable — Regions VI, VII, IX, X, XI (Davao), XII (Soccsksargen), ARMM, and NCR showed structurally weak supply chains and low supply stability. Notably, NCR is classified as Vulnerable despite being the country's economic center, reflecting its complete dependence on external food sources.
- Secure — Regions I, II, IV, V, VIII, XIII, and CAR demonstrated moderate price volatility and resilient food systems, making them best positioned to withstand market disruptions.

4. Price Prediction — Baseline vs. Fine-Tuned Random Forest



Metric	Baseline RF	Fine-Tuned RF
MAE	**₱ 8.00**	**₱ 2.29**
RMSE	**₱ 16.04**	**₱ 8.64**

The Fine-Tuned Random Forest (n_est=200, depth=12, leaf=1) significantly outperformed the baseline model, achieving a 71.4% reduction in MAE and a 46.1% reduction in RMSE. By applying 3× sample weights to high-price spike events (top 5th percentile), the model became substantially more responsive to typhoon-driven price surges. As shown in the figure above, the fine-tuned model's predicted prices track the actual price curve far more closely across the entire January–July 2021 test period compared to the baseline, which consistently lagged during sharp price transitions.

These findings provide clear, data-driven guidance for policymakers to prioritize food security interventions in high-risk and vulnerable regions before price crises escalate.

Challenges and Solutions

- **Challenge:** Training on the full 2.5M rows led to kernel crashes and excessive training time.
- **Solution:** Implemented a sample_size = 700,000 strategy for the training phase to ensure model stability.

Conclusion

The project successfully developed a robust predictive and analytical framework for monitoring staple food prices across 17 regions in the Philippines. By processing over two decades of historical data (2000–2021), the team demonstrated that machine learning, specifically the Random Forest Regressor, is a highly effective tool for anticipating market fluctuations.

Key Takeaways:

- **Predictive Accuracy:** The implementation of a fine-tuned Random Forest model achieved a Mean Absolute Error (MAE) of approximately 8.00, indicating that the model can forecast prices with high reliability even amidst the inherent volatility of the agricultural market.
- **Driving Factors:** The analysis revealed that price momentum (7-day and 14-day lags) and environmental variables (14-day and 30-day rolling precipitation/temperature) are critical predictors. This confirms that food security in the Philippines is deeply intertwined with both market behavior and climate patterns.
- **Risk Identification:** The regional risk classification successfully categorized regions like Region XI (Davao) and Region XII (Soccsksargen) as "Vulnerable." This data-driven categorization provides actionable insights for policymakers to prioritize food price stabilization efforts and resource allocation in these specific areas.
- **Scalability:** Despite the computational challenges of handling 2.5 million records, the team successfully optimized the pipeline using strategic data sampling and efficient feature encoding, proving that the system is scalable for future real-time monitoring applications.

In summary, this project met all its primary objectives. It not only provides a high-performing model for price forecasting but also establishes a data-driven foundation for addressing regional food insecurity and price instability in the Philippines.